



**DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES  
E INFORMÁTICA**

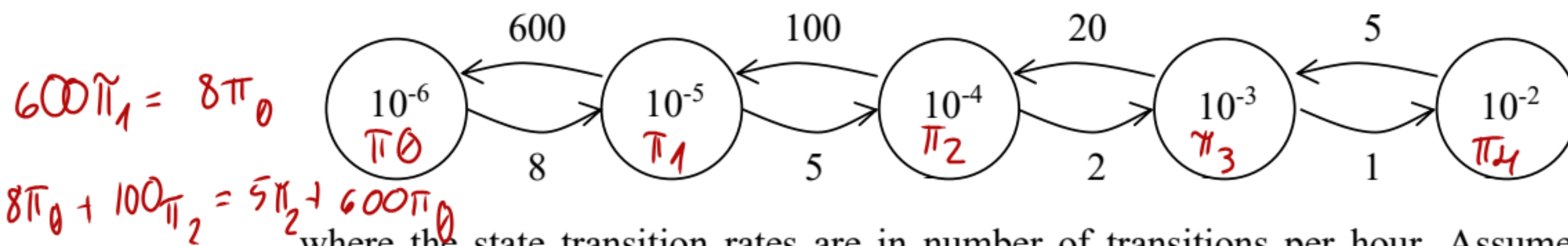
**LICENCIATURA EM ENGENHARIA DE COMPUTADORES E INFORMÁTICA**

**ANO 2025/2026**

**MÉTODOS PROBABILÍSTICOS PARA ENGENHARIA DE  
COMPUTADORES E INFORMÁTICA**

**PRACTICAL GUIDE No. 4**

- Consider a wireless link between multiple stations for data communications. Consider the bit error rate (*ber*) introduced by the wireless link (due to the variation of the propagation and interference factors along with time) given by the following Markov chain:



where the state transition rates are in number of transitions per hour. Assume that all stations detect with a probability of 100% when the data frames sent by the other stations are received with errors.

- Determine the probability of the link being on each state. *- em cadeias de Markov em tempo contínuo é a mesma coisa*
- Determine the average time percentage the link is on each state.
- Determine the average holding time (in minutes) of the link on each state.
- Determine the probability (in %) of a data frame of 100 Bytes to be received with errors (i.e., with at least one bit in error). *'800 bits*
- Determine the probability (in %) of a data frame of 1500 Bytes to be received without errors.
- When a data frame of 1500 Bytes is received with errors, what is the probability of the link being in state  $10^{-6}$ ? And in state  $10^{-2}$ ? Compare these probabilities with the probabilities in 1(a) and draw conclusions.

- Consider a company providing all services on a single server. The server can break down due to 2 causes: **hardware failure** or **software failure**. The amount of time the server runs until a hardware/software failure is exponentially distributed with an average of **500 days** and **90 days**, respectively. The hardware/software repair time is exponentially distributed with an average of **1 day** and **4 hours**, respectively. A hardware failure **requires first the hardware repair and then the software repair**, while a software failure requires only the software repair.

- Draw the state transition diagram of the Markov chain modeling this system with state transition rates in number of transitions per day.
- Define, in Matlab, the corresponding state transition matrix.
- Compute the limiting probabilities (in %) of each state.
- What is the probability (in %) of the server being available?
- What is the average time (in hours) that the server is down in one month of 30 days?

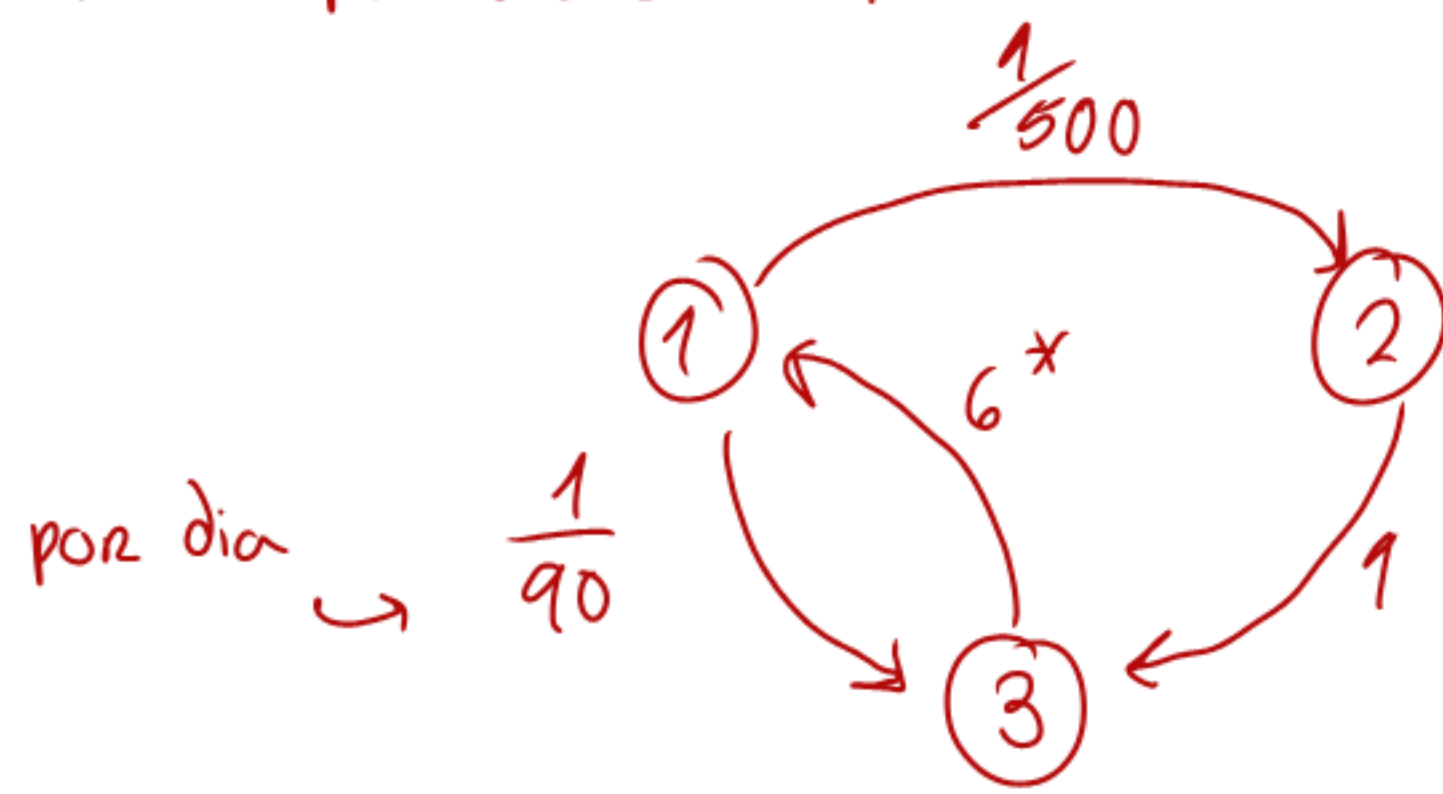
- Consider a Data Center with **100 servers** and  $t$  maintenance technicians. The amount of time each server runs until it fails is exponentially distributed with an average of **200 days**. The server repair time of each technician is exponentially distributed with an average of **36 hours**. Consider a birth-death process with states  $n = 0, 1, \dots, 100$  representing the number of failing servers.

- Consider  $t = 1$  technician. Compute first the state limiting probabilities and then:
  - determine the average number of servers down,  *$0 \times p(0) +$*
  - determine the average number of technicians in repairing activities,
  - determine the probability (in %) of at least 90 servers being in operation.

*$p(0 \rightarrow 10)$*

2.a) 1- server funcionando 2- falha hardware 3- falha software

$$\neq \frac{1}{4} \times 24 = 6$$

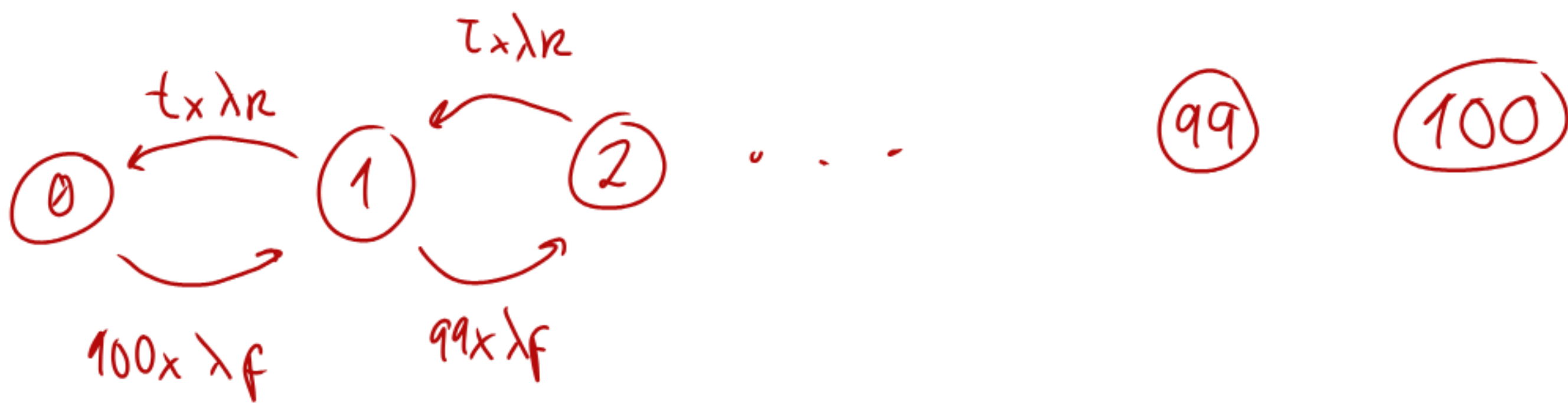


não é processo nascimento e morte!

3)

$$\lambda_f = \frac{1}{200}$$

$$\lambda_r = \frac{24}{36}$$



a)  $t=1$

- (b) Repeat 3(a) considering  $t = 2$  technicians.
  - (c) Repeat 3(a) considering  $t = 3$  technicians. Compare all results and draw conclusions.
4. Consider a  $M/M/1$  queuing system such that the arriving rate of clients is  $\lambda = 8$  clients per minute and the average serving time of each client is  $1/\mu = 0.1$  minutes.
- (a) What is the average time each client stays in the system?
  - (b) What is the average server occupation (in %)?
  - (c) What is the average queue occupation, i.e., the average number of clients in the queue?
  - (d) What is the probability (in %) of the queue occupation becoming larger than 6 clients? And larger than 20 clients? And larger than 40 clients?
  - (e) Using Matlab *plot* function, plot a graph of the average time each client stays in the system as a function of the client arriving rate  $\lambda$  (from  $\lambda = 1$  to  $\lambda = 9$ ).
5. Consider a  $M/M/1/m$  queuing system such that the arriving rate of clients is  $\lambda = 80$  clients per second, the average serving time of each client is  $1/\mu = 0.01$  seconds and the system capacity is  $m = 10$  clients. Clients are discarded when they arrive and the system is full.
- (a) What is the probability of each client being discarded?
  - (b) What is the average time (in milliseconds) each client stays in the system when it is not discarded?
  - (c) What is the average queue occupation?
  - (d) Using Matlab *bar* function, plot a bar chart of the client discard probability as a function of the system capacity  $m$  (for  $m = 5, 10, 15, \dots, 40$ ). What do you conclude from this chart?